UNIVERSITÀ DI PISA

Department of Computer Science

Master programme in Data Science and Business Informatics

Data Mining: Foundations

Project Report

Board Games Dataset

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Table of Contents

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1. Data Understanding & Preparation

1.1 Data Semantics

The dataset variables are organized into distinct functional groups: game metadata (identifiers, names, descriptions), physical characteristics (player counts, playtime, age recommendations), complexity metrics (GameWeight, ComWeight, LanguageEase), community engagement indicators (ownership counts, wishlist entries, user ratings), publication metadata (year, manufacturer specifications), ranking positions across eight game categories, and binary category flags indicating game classification. This grouping facilitates targeted analysis where engagement metrics serve as primary targets, complexity variables function as predictors, and ranking/category fields enable stratified analysis.

Variable	Type	Meaning	Range	Notes
BGGId	Integer	Unique BoardGameGeek identifier	1-330154	Primary key, exclude from analysis
YearPublished	Integer	Year of game publication	-3500 to 2021	Negative = ancient games (Chess, Go)
GameWeight	Float	Official complexity rating	0.0 - 5.0	0 = not rated (506 cases)
MinPlayers	Integer	Minimum players required	0 - 10	0 = unknown (50 cases)
MaxPlayers	Integer	Maximum players supported	0 - 999	999 = unlimited; 0 = unknown
NumOwned	Integer	Count of users owning the game	0 - 166,497	Heavy right skew, needs log transform
NumUserRatings	Integer	Count of user ratings received	30 - 108,101	Min 30 threshold for inclusion
MfgPlaytime	Integer	Manufacturer playtime (minutes)	0 - 60,000	0 = not specified (780 cases)
ComAgeRec	Float	Community age recommendation	2.0 - 21.0	25.2% missing values
Family	String	Game family/series name	Categorical	69.6% missing; optional grouping
Rank:* columns	Integer	Ranking in 8 game categories	1 - 21,925	Lower = better rank
Cat:* columns	Binary	Category membership flags	0 or 1	8 binary flags; not mutually exclusive

Table 1.1: Data Dictionary for Key Variables

Interpretation: The table presents 12 key variables selected for their analytical relevance. Numeric engagement variables (NumOwned, NumUserRatings) exhibit extreme skewness requiring transformation. Placeholder values (0, 999) serve as sentinels for missing or unlimited data.

Variable Usage Classification:

Numeric analysis: GameWeight, MinPlayers, MaxPlayers, NumOwned, NumWant, NumWish, NumUserRatings, MfgPlaytime, ComAgeRec, YearPublished. These continuous/discrete variables support correlation analysis, clustering, and regression modeling.

Careful treatment required: Rank columns contain ordinal data with interpretation complexity (lower=better); Category binary flags (Cat:*) are suitable for stratification and classification targets.

Excluded from modeling: BGGId (identifier), Name (text), Description (long text), ImagePath (URL), GoodPlayers (complex string), Rating (string requiring parsing). These provide reference but not analytical value.

1.2 Distribution of Variables and Descriptive Statistics

1.2.1 Univariate Numeric Summaries

The following 10 numeric variables were selected based on their analytical importance: YearPublished captures temporal distribution, GameWeight and ComAgeRec represent complexity attributes, MinPlayers and MaxPlayers define game format, and the engagement metrics (NumOwned, NumWant, NumWish, NumUserRatings) quantify community interest. MfgPlaytime characterizes session length.

Variable	Count	Miss %	Mean	Median	Std	Min	Max	Skew
YearPublished	21,925	0.0	1985.5	2011	212.5	-3500	2021	-11.32
GameWeight	21,925	0.0	1.98	1.97	0.85	0.0	5.0	0.40
MinPlayers	21,925	0.0	2.01	2	0.69	0	10	1.70
MaxPlayers	21,925	0.0	5.71	4	15.01	0	999	42.39
ComAgeRec	16,395	25.2	10.0	10	3.27	2	21	0.14
NumOwned	21,925	0.0	1,468	320	5,294	0	166,497	12.52
NumWant	21,925	0.0	41.7	9	117.3	0	2,031	6.96
NumWish	21,925	0.0	228.5	39	788.5	0	19,182	9.35
NumUserRatings	21,925	0.0	862	123	3,639	30	108,101	12.59
MfgPlaytime	21,925	0.0	90.5	45	529.7	0	60,000	74.74

Table 1.2: Descriptive Statistics for Key Numeric Variables

Key Takeaways: (1) Engagement variables exhibit extreme right skew (NumOwned skew=12.52, NumUserRatings skew=12.59), indicating most games have low engagement while few achieve viral popularity. (2) YearPublished shows a strong negative skew (-11.32) due to ancient games like chess (3500 BC). (3) MaxPlayers contains an implausible maximum of 999, a clear placeholder for unlimited players. (4) ComAgeRec has a substantial missing rate (25.2%), requiring imputation or exclusion decisions. (5) MfgPlaytime extremes (60,000 minutes = 1000 hours) suggest data entry errors or campaign-style games.

1.2.2 Univariate Plots

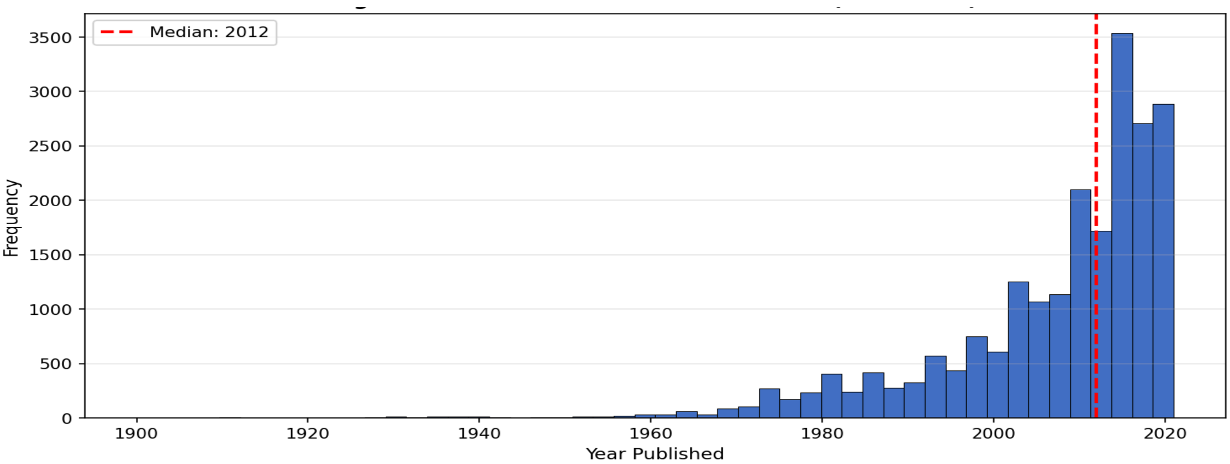

Fig 1.1: Distribution of year publication variable.

Figure 1.1 shows the distribution of publication years filtered to 1900-2021. The distribution reveals a pronounced right-skew toward recent decades with a median at 2011. The exponential growth in board game publication post-2000 reflects the modern board game renaissance, which has significant implications for temporal bias in any predictive modelling. Historical games (pre-1950) represent a sparse but culturally significant segment.

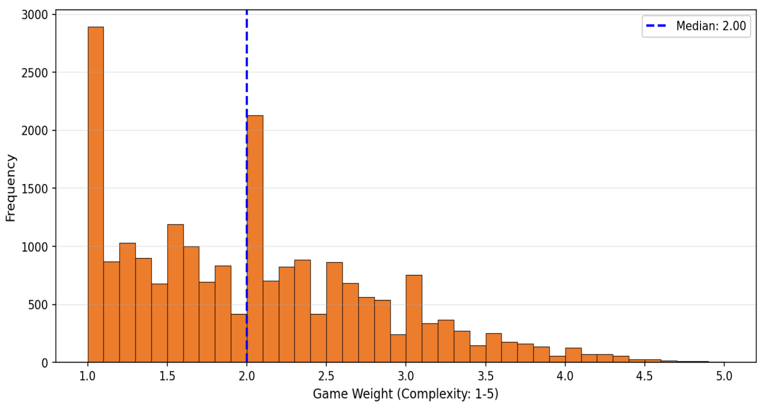

Fig 1.2: Distribution of game complexity variable.

Figure 1.2 displays the game complexity weight distribution (excluding zeros). The near-symmetric distribution centered around 2.0 (medium complexity) indicates that BGG's community-rated games span the full complexity spectrum with slight preference for accessible games. The bounded 1-5 scale requires no transformation, making GameWeight directly suitable for clustering and regression.

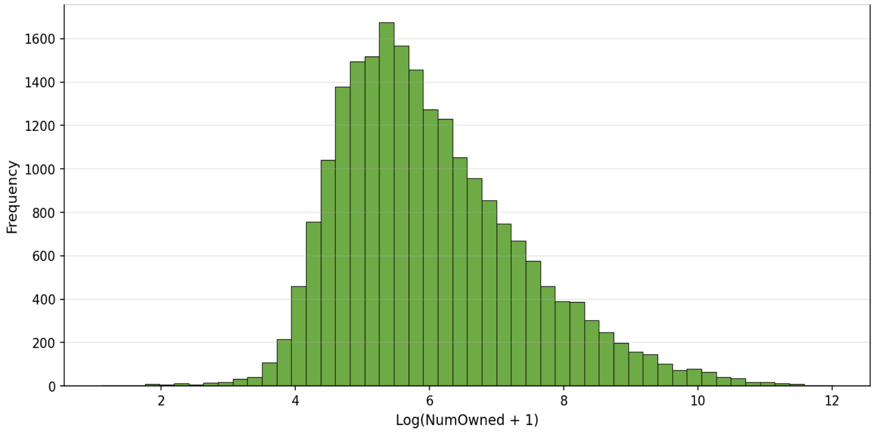

Fig 1.3: Distribution of owner ship count variable

Figure 1.3 presents NumOwned after log1p transformation. The transformed distribution approximates normality, validating log transformation as essential preprocessing for engagement metrics. The original variable's skewness of 12.52 is reduced substantially, enabling effective use in distance-based clustering and linear regression models.

1.3 Pairwise Exploration

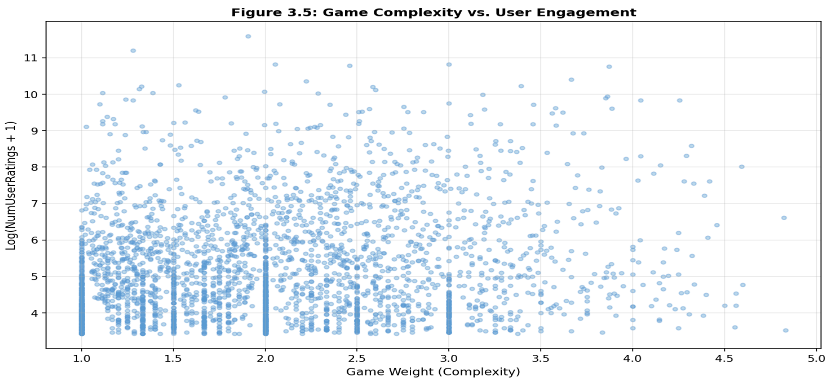

Fig: 1.4: Game complexity vs User Engagement

Figure 1.4 examines the relationship between GameWeight and log-transformed NumUserRatings. The scatter plot reveals a moderate positive correlation suggesting that more complex games tend to attract more engaged audiences who rate games. However, the relationship is non-linear with substantial variance, indicating that complexity alone does not determine engagement. Games across all complexity levels can achieve high engagement, but the ceiling increases with complexity.

Additional Pairwise Relationships:

NumOwned vs NumUserRatings (r=0.985): Near-perfect correlation indicates these metrics capture the same underlying popularity construct. One variable should be eliminated to avoid multicollinearity in regression.

GameWeight vs ComWeight (r=0.997): Virtual identity between official and community complexity ratings validates data quality but necessitates removing one as redundant.

Implications for Later Modules: The high correlations among engagement metrics suggest dimensionality reduction via variable elimination or PCA. Classification tasks should use non-redundant predictors. Clustering may benefit from log-transformed engagement variables to prevent dominance by high-magnitude features.

1.4 Assessing Data Quality and Variable Transformations

1.4.1 Missing Values

Variable	Missing %	Decision	Justification
Family	69.61%	Exclude from modeling	Majority missing; optional metadata for game series
LanguageEase	26.87%	Exclude or impute	High missingness; use median if needed

Variable	Missing %	Decision	Justification
ComAgeRec	25.22%	Impute with median	Valuable for analysis; symmetric distribution supports median imputation
ImagePath	0.08%	Keep (excluded anyway)	Negligible; URL field excluded from analysis
Description	0.00%	Exclude from numeric analysis	Text field; could support NLP in future work

Table 1.3: Missing Value Analysis and Handling Decisions

Missing Data Policy: Variables exceeding 50% missingness (Family) are excluded from modeling. Variables with 20-50% missingness (LanguageEase, ComAgeRec) receive median imputation when the distribution is approximately symmetric, with a binary indicator created for models sensitive to imputation. Variables below 1% missingness are retained as-is.

1.4.2 Errors and Semantic Inconsistencies

Placeholder and Sentinel Values: Several numeric fields use zero or extreme values as placeholders:

MaxPlayers = 999 (3 cases): Indicates unlimited player count. Decision: Cap at reasonable maximum (e.g., 20) or treat as separate category.

MaxPlayers = 0 (173 cases), MinPlayers = 0 (50 cases): Missing data encoded as zero. Decision: Replace with median player counts.

GameWeight = 0 (506 cases): Indicates unrated games. Decision: Exclude from complexity-based analysis or impute with category median.

MfgPlaytime = 0 (780 cases), MfgAgeRec = 0 (1,325 cases): Missing manufacturer data. Decision: Replace with community equivalent when available.

Implausible Values: YearPublished ranges from -3500 to 2021. Negative years represent ancient games (Chess, Go, Senet). These are legitimate historical entries and are retained. MfgPlaytime maximum of 60,000 minutes (1,000 hours) likely represents campaign-style games or data entry errors; values exceeding 1,440 minutes (24 hours) are capped.

Duplicates: Analysis confirmed zero duplicate BGGId values and zero fully duplicate rows, indicating good data integrity from the source.

1.4.3 Outliers

Outlier Handling Policy: Domain-informed logical rules are applied rather than purely statistical methods (IQR) because legitimate extreme values exist in this dataset. Board games exhibit natural variation where blockbuster titles (Catan, Pandemic) have genuinely extreme engagement metrics.

Examples and Actions:

NumOwned outliers (13% by IQR): Retained. High ownership reflects genuine popularity (e.g., Catan with 166,497 owners). Log transformation normalizes the distribution without data loss.

MaxPlayers = 999: Capped at 20. Three games with 999 represent unlimited-player party games; cap preserves information while preventing distortion.

MfgPlaytime > 1440 min: Capped at 1440 (24 hours). Values beyond represent campaign games or errors; capping maintains analytical utility.

1.4.4 Transformations for Later Modules

Log1p Transformation: Applied to heavy-tailed engagement counts (NumOwned, NumWant, NumWish, NumUserRatings, NumExpansions) to reduce skewness and enable effective distance-based clustering. The log1p function handles zero values gracefully.

Standardization (Z-score): Required for clustering algorithms (K-means, hierarchical) that use Euclidean distance. Applied after log transformation to engagement variables and directly to bounded variables (GameWeight, player counts).

Encoding Strategy: The eight Cat:* binary columns require no encoding. Rank columns could be converted to percentile ranks for comparability. Family categorical (when used) would require one-hot encoding or target encoding depending on cardinality analysis.

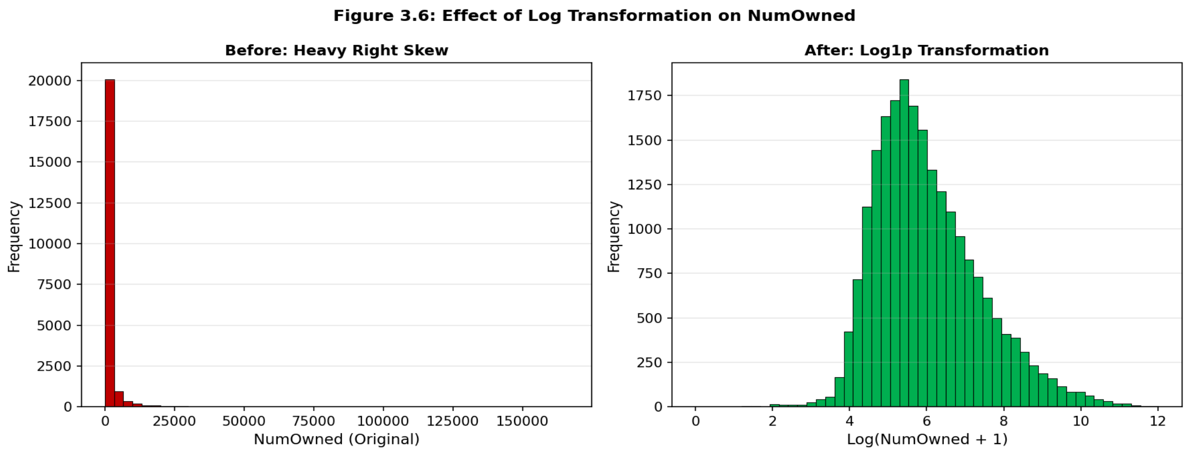

Fig 1.5: Effect of Log Transformation on NumOwned

Figure 1.5 demonstrates the effect of log1p transformation on NumOwned. The original distribution (left) shows extreme right skew with most values near zero and a long tail extending to 166,497. After transformation (right), the distribution approaches normality, validating log transformation as essential preprocessing for distance-based methods and linear models.

1.5 Pairwise Correlations and Variable Elimination

Correlation analysis was conducted on 12 key numeric variables selected based on analytical relevance: temporal (YearPublished), complexity (GameWeight), player configuration (MinPlayers, MaxPlayers), engagement metrics (NumOwned, NumWant, NumWish, NumUserRatings, NumWeightVotes), and playtime variables (MfgPlaytime, ComMinPlaytime).

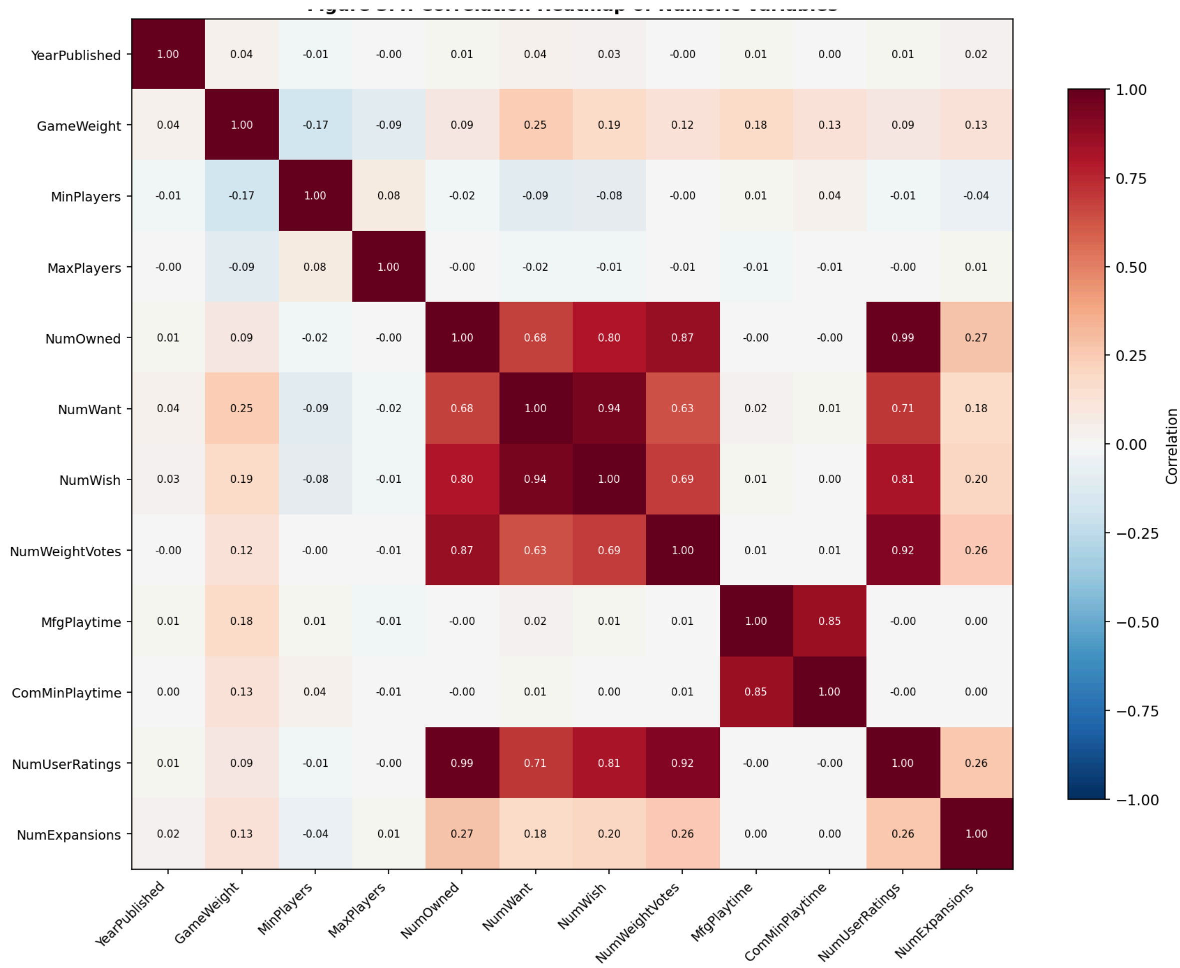

Figure 1.6 displays the correlation heatmap revealing several highly correlated variable clusters. The engagement block (NumOwned, NumWish, NumUserRatings, NumWeightVotes) shows correlations exceeding 0.80, indicating these capture related popularity constructs. The playtime cluster (MfgPlaytime, ComMinPlaytime) also exhibits near-perfect correlation (r=0.86). YearPublished shows weak negative correlations with engagement metrics, suggesting modern games may have accumulated fewer ratings due to shorter availability.

Variable A	Variable B	Correlation	Decision	Justification
NumOwned	NumUserRatings	0.985	Keep NumOwned	More interpretable metric
NumWant	NumWish	0.940	Keep NumWish	Larger scale, more variance
NumWeightVotes	NumUserRatings	0.917	Drop NumWeightVotes	Redundant with ratings
MfgPlaytime	ComMinPlaytime	0.855	Keep MfgPlaytime	Official source preferred

Table 1.4: Highly Correlated Variable Pairs (|r| > 0.85)

Redundancy Elimination: Based on correlations exceeding 0.85, the following variables are dropped to prevent multicollinearity: NumUserRatings (redundant with NumOwned), NumWant (redundant with NumWish), NumWeightVotes (redundant with engagement metrics), ComMinPlaytime and ComMaxPlaytime (redundant with MfgPlaytime), and ComWeight (near-identical to GameWeight). This reduces the feature set while preserving information content.

1.6 Output of Preparation

Final Dataset Characteristics:

Variables Retained: YearPublished, GameWeight, MinPlayers, MaxPlayers, NumOwned, NumWish, MfgPlaytime, ComAgeRec (imputed), NumExpansions, all Cat:* binary flags (8), and selected Rank:* columns for stratified analysis.

Variables Excluded: BGGId, Name, Description, ImagePath (non-analytical); Family (69% missing); NumUserRatings, NumWant, NumWeightVotes, ComWeight, ComMinPlaytime, ComMaxPlaytime (redundant); GoodPlayers, Rating (complex strings).

Missing Value Handling: ComAgeRec imputed with median (10.0); LanguageEase excluded due to high missingness (26.9%); sentinel values (0 for players, weight, playtime) replaced with appropriate medians or category-specific values.

Outlier Treatment: MaxPlayers capped at 20; MfgPlaytime capped at 1440; engagement outliers retained after log transformation.

Transformations Applied: Log1p to NumOwned, NumWish, NumExpansions for skew reduction; z-score standardization for clustering; binary flags require no transformation.

Remaining Limitations: Temporal bias toward recent games may affect temporal analysis; engagement metrics favor established games over newer releases; category assignments are not mutually exclusive, creating potential classification ambiguity; ancient games (negative years) require careful handling of year-based features.

2. Clustering Analysis

This chapter presents unsupervised clustering analysis of the board games dataset to discover natural groupings based on gameplay characteristics, complexity metrics, and community engagement patterns. Three clustering approaches are employed: K-Means (centroid-based), DBSCAN (density-based), and Hierarchical Agglomerative Clustering. The analysis uses 26 features from the prepared dataset including log-transformed popularity metrics, standardized complexity scores, and binary category flags. All features were z-score normalized prior to clustering to ensure equal contribution to distance calculations.

2.1 K-Means Clustering

2.1.1 Parameter Selection

K-Means clustering requires specification of the number of clusters k. Two complementary methods were employed to identify the optimal value: the Elbow method examining within-cluster sum of squares (inertia) and Silhouette analysis measuring cluster cohesion and separation.

The Elbow plot (Figure 2.1, left) shows inertia decreasing monotonically from 514,639 at k=2 to 298,567 at k=12. No clear elbow point is visible, which is characteristic of datasets with overlapping clusters in high-dimensional space. The Silhouette plot (Figure 2.1, right) provides more decisive guidance, showing a clear peak at k=11 with a silhouette score of 0.234. Secondary peaks occur at k=7 (0.187) and k=6 (0.180), but k=11 maximizes cluster quality.

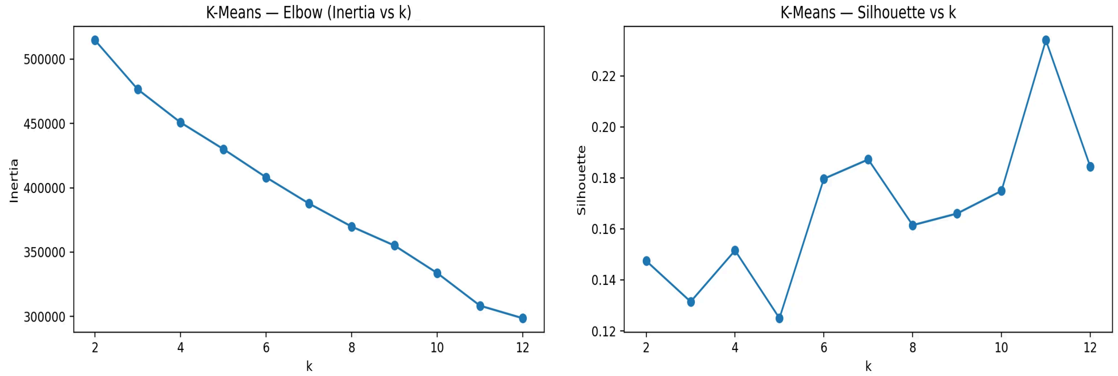

Figure 2.1: K-Means parameter selection. Left: Elbow plot (Inertia vs. k). Right: Silhouette score vs. k showing optimal k=11.

Table 2.1: K-Means Grid Search Results

k	Inertia	Silhouette	Davies-Bouldin	Calinski-H.
2	514,639	0.147	2.726	2,360
6	408,089	0.180	1.862	1,740
7	387,781	0.187	1.638	1,717
11	308,317	0.234	1.339	1,860
12	298,567	0.184	1.499	1,811

The selected configuration (k=11, highlighted) achieves the best silhouette score (0.234), lowest Davies-Bouldin index (1.339, indicating better cluster separation), and competitive Calinski-Harabasz score (1,860). The combination of metrics confirms k=11 as the optimal choice.

2.1.2 Cluster Distribution and Profiles

The K-Means solution with k=11 produces clusters of varying sizes, ranging from 212 games (Cluster 10, 1.0%) to 10,020 games (Cluster 2, 45.7%). The largest cluster (Cluster 2) represents generic games without strong distinguishing characteristics, while smaller clusters capture specialized game types.

Table 2.2: K-Means Cluster Size Distribution

Cluster	Count	Percentage	Primary Characteristic
	10,020	45.7%	Generic/Low engagement
	3,243	14.8%	War games
	2,622	12.0%	Strategy games
	1,894	8.6%	Family games
	994	4.5%	Abstract games
	841	3.8%	Children's games
	632	2.9%	Party games
	303	1.4%	Card games (CGS)
	212	1.0%	Historical/Ancient

Cluster profiling reveals semantically meaningful groupings. Cluster 0 (Family games) shows Cat:Family=0.998, significantly above the global mean of 0.106, with moderate complexity (GameWeight=1.70) and high ownership. Cluster 5 (War games) exhibits Cat:War=0.961, extended playtime (239 min vs. global 80 min), and high complexity (2.91). Cluster 4 (Party games) shows Cat:Party=1.0 with higher minimum player counts (2.8 vs. 2.0) and maximum players (11.8 vs. 5.6). Cluster 10 captures ancient games with mean YearPublished of -82 (representing games like Chess and Go), higher abstraction tendency (Cat:Abstract=0.20), and lower community engagement due to limited digital tracking.

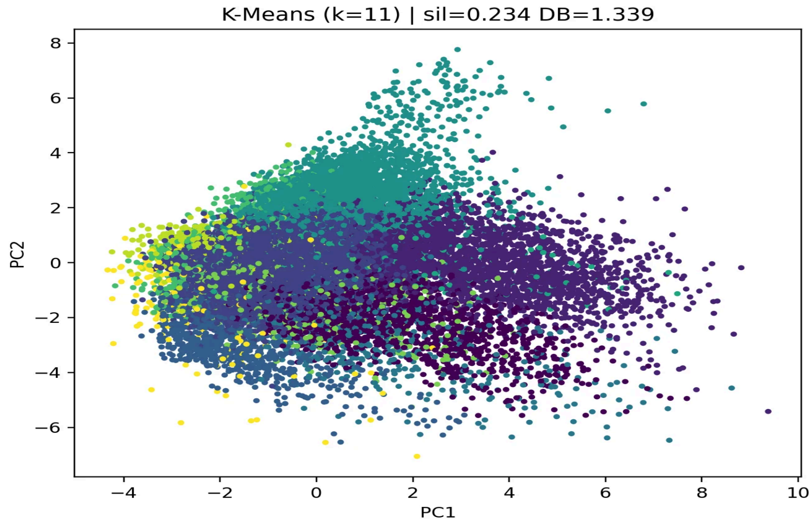

Figure 2.2: PCA visualization of K-Means clusters (k=11). Silhouette=0.234, Davies-Bouldin=1.339.

The PCA projection (Figure 2.2) shows moderate cluster separation in the reduced two-dimensional space. While some overlap exists in the central region (expected given the continuous nature of board game characteristics), distinct clusters emerge at the periphery corresponding to specialized game categories. The upper-left region contains high-engagement strategy games, while the lower regions show family and children's games with lower complexity.

2.2 DBSCAN Clustering

2.2.1 Parameter Selection

DBSCAN requires two parameters: epsilon (ε, neighborhood radius) and min_samples (minimum points to form a dense region). The k-distance plot (Figure 3.3) with k=min_samples=10 shows a gradual increase in distances with a sharp upturn around the 20,000th point, suggesting ε values between 1.5 and 3.0 as reasonable candidates.

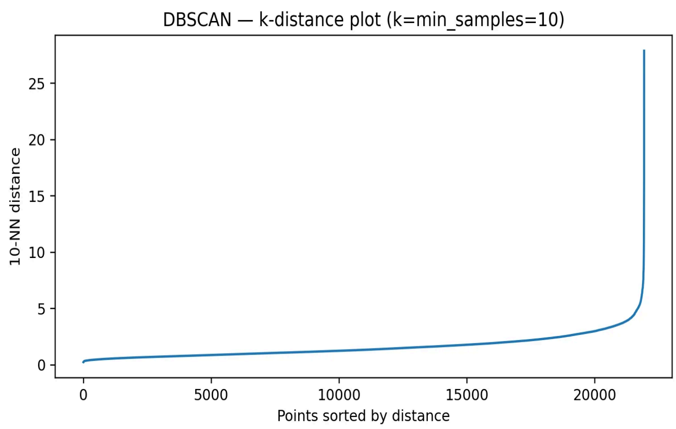

Figure 2.3: k-distance plot for DBSCAN parameter selection (k=min_samples=10).

A grid search was conducted over ε ∈ {0.3, 0.5, ..., 2.5} and min_samples ∈ {5, 10, 20}. The results reveal a trade-off between cluster granularity and noise fraction. Lower min_samples values produce more clusters but with higher fragmentation. The configuration ε=2.5, min_samples=20 was selected, achieving silhouette=0.231 with 42 clusters and 12.1% noise, balancing meaningful cluster discovery with reasonable noise levels.

Table 2.3: DBSCAN Grid Search Results (Selected Configurations)

ε	min_samples	Clusters	Noise %	Silhouette	DB Index
1.7	5	141	18.8%	0.186	1.17
2.1	10	72	14.6%	0.215	1.27
2.5	20	42	12.1%	0.231	1.32
1.7	20	50	31.2%	0.221	1.21

2.2.2 Cluster Analysis

DBSCAN identifies 42 clusters plus a noise cluster (-1) containing 2,663 points (12.1%). The noise cluster captures outlier games with unusual combinations of features, including games with exceptionally high engagement (BestPlayers z-score=0.90), numerous expansions, or unusual category combinations. The largest cluster (Cluster 0) contains 7,151 games (32.6%) characterized by low engagement and absence of distinctive category membership—essentially the 'generic game' cluster.

Notable specialized clusters include: Cluster 1 (War games, n=2,387) with Cat:War=1.0, high complexity (2.81), and extended playtime (190 min); Cluster 2 (Family games, n=1,006) with Cat:Family=1.0 and moderate complexity; Cluster 3 (Children's games, n=611) with Cat:Childrens=1.0, low complexity (1.16), and young age recommendations (ComAgeRec=5.6); Cluster 5 (Kickstarted strategy games, n=295) showing both Kickstarted=1.0 and Cat:Strategy=1.0 with high engagement metrics; and Cluster 37 (Heavy war games, n=53) with extreme playtime (1,365 min average) and maximum complexity (3.91).

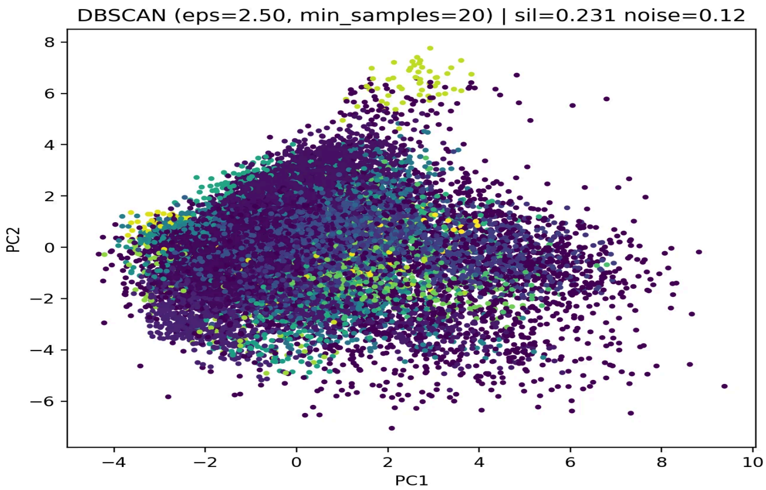

Figure 2.4: PCA visualization of DBSCAN clusters (ε=2.5, min_samples=20). Silhouette=0.231, Noise=12.1%.

The PCA projection (Figure 2.4) reveals that DBSCAN successfully identifies dense regions corresponding to game categories while appropriately labeling sparse boundary points as noise. The density-based approach captures naturally varying cluster shapes better than the spherical assumption of K-Means, though the high dimensionality limits visual separation in two dimensions.

2.3 Hierarchical Agglomerative Clustering

2.3.1 Dendrogram Analysis

Hierarchical clustering with Ward's linkage was applied to a sample of 1,500 games (due to computational constraints of the $O(n^2)$ algorithm). Ward's method minimizes within-cluster variance at each merge, producing compact, spherical clusters suitable for this mixed-feature dataset.

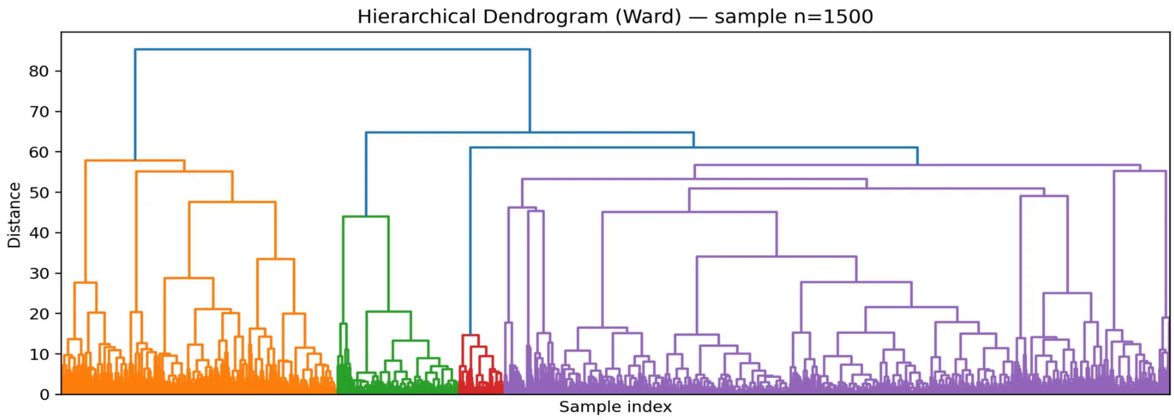

Figure 2.5: Hierarchical dendrogram (Ward linkage, n=1,500 sample). Color-coded branches indicate natural cluster boundaries.

The dendrogram (Figure 2.5) shows the hierarchical structure of game relationships. A major split occurs at distance ~85, dividing games into two primary groups: mainstream games (orange branch, left) and specialized/complex games (purple branch, right). Within the mainstream branch, further divisions separate family-oriented games (left sub-branch) from generic low-engagement games. The specialized branch shows greater internal heterogeneity with multiple sub-clusters emerging at lower distance thresholds.

2.3.2 Optimal k Selection

Grid search over k ∈ {2, ..., 12} reveals consistently improving silhouette scores with increasing k, from 0.245 at k=2 to 0.230 at k=12. The best silhouette (0.245) occurs at k=2, reflecting the strong binary split visible in the dendrogram. However, this coarse partition sacrifices interpretability by merging semantically distinct game types.

Table 2.4: Hierarchical Clustering Grid Search Results

k	Silhouette	Davies-Bouldin	Calinski-Harabasz
2	0.245	2.580	1,747
5	0.135	1.800	1,527
8	0.184	1.647	1,574
11	0.219	1.305	1,669
12	0.230	1.259	1,717

The selected k=2 solution produces highly imbalanced clusters: Cluster 0 contains 18,825 games (85.9%) and Cluster 1 contains 3,100 games (14.1%). Cluster 1 (minority) is characterized by high engagement (BestPlayers z-diff=1.51, NumOwned z-diff=1.34), thematic content (Cat:Thematic z-diff=1.44), and party game features (Cat:Party z-diff=1.04). This cluster represents 'hit games' with strong community presence, while Cluster 0 captures the long tail of less popular titles.

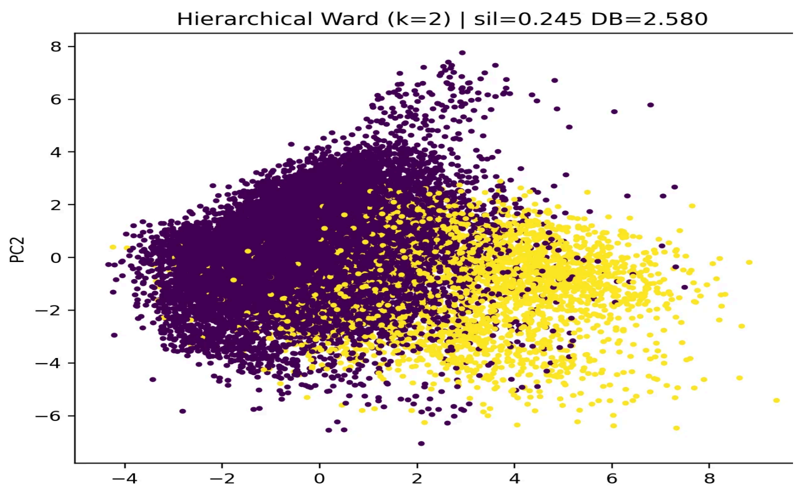

Figure 2.6: PCA visualization of Hierarchical clusters (k=2, Ward). Silhouette=0.245, Davies-Bouldin=2.580.

2.4 Algorithm Comparison and Discussion

Table 2.5: Clustering Algorithm Comparison Summary

Algorithm	Clusters	Silhouette	DB Index	Noise %
K-Means (k=11)	11	0.234	1.339	0%
DBSCAN	42	0.231	1.317	12.1%
Hierarchical (k=2)	2	0.245	2.580	0%

All three algorithms achieve similar silhouette scores (0.23-0.25), indicating comparable overall cluster quality. However, they differ substantially in interpretability and use-case suitability:

K-Means (Recommended): Provides the best balance between granularity (11 clusters) and interpretability. Each cluster corresponds to recognizable game categories (family, war, party, children's, abstract, etc.), enabling practical applications such as recommendation systems or market segmentation. The lowest Davies-Bouldin index (1.339) indicates well-separated clusters.

DBSCAN: Identifies more granular structure (42 clusters) and appropriately handles outliers through noise classification. Useful for anomaly detection and discovering niche game categories. However, the large number of clusters complicates interpretation, and the 12.1% noise fraction may exclude games of interest.

Hierarchical: Reveals the fundamental structure of the dataset through the dendrogram, showing that games primarily divide into 'popular/engaging' versus 'generic/niche' categories. The k=2 solution, while statistically optimal, is too coarse for most practical applications but provides valuable insight into the primary axis of variation in the data.

The moderate silhouette scores across all methods (~0.23-0.25) reflect the inherent overlap in board game characteristics—games often span multiple categories and exhibit continuous rather than discrete variation in complexity and popularity. This is consistent with domain knowledge: board games exist on spectra of complexity, player count, and thematic content rather than in well-separated natural clusters.

Conclusion: K-Means with k=11 is recommended as the primary clustering solution for this dataset. It achieves competitive quantitative metrics while producing semantically meaningful clusters aligned with established board game categories, making it suitable for downstream applications including game recommendation, market analysis, and feature engineering for supervised learning tasks.

3. Classification and Regression Analysis

This report presents the results of classification and regression tasks performed on the Board Games dataset containing 21,925 records with 29 features (18 numeric and 11 binary). The primary classification task predicts the Rating variable (categorized as low, medium, or high), while the regression task predicts NumOwned as a continuous target.

3.1 Target Variable Definition

The Rating variable was transformed into three ordinal categories: low (1,449 samples), medium (1,929 samples), and high (1,007 samples) in the test set. This distribution shows a moderate class imbalance with the medium category being most prevalent.

3.1.1 Models Evaluated and Hyperparameter Tuning

Three classification algorithms were evaluated using grid search with cross-validation: Decision Tree (tuned ccp_alpha, criterion, max_depth, and min_samples_leaf parameters), K-Nearest Neighbors (tuned n_neighbors, weights, metric, and p parameters), and Naive Bayes (tuned var_smoothing parameter).

3.1.2 Model Performance Comparison

Table 3.1: Classification Model Comparison

Model	Best Parameters	Accuracy	F1 Macro	F1 Weighted
Decision Tree	ccp_alpha=0.001, entropy	0.686	0.688	0.686

KNN	k=21, distance, minkowski	0.669	0.665	0.668
Naive Bayes	var_smoothing=1e-9	0.506	0.474	0.435

The Decision Tree classifier achieved the best performance with 68.6% accuracy and F1-macro of 0.688. K-Nearest Neighbors performed comparably (66.9% accuracy), while Naive Bayes showed significantly lower performance (50.6%), suggesting the feature independence assumption does not hold for this dataset.

3.1.3 Detailed Classification Metrics

Table 3.2: Per-Class Performance Metrics (Decision Tree)

Class	Precision	Recall	F1-Score	Support
High	0.724	0.643	0.681	1,007
Low	0.708	0.745	0.726	1,449
Medium	0.652	0.665	0.658	1,929

The per-class analysis reveals that the high rating class achieves the highest precision (0.724) but lowest recall (0.643), indicating the model is conservative in predicting high ratings but accurate when it does. The low class shows the most balanced precision-recall trade-off (0.708/0.745). The medium class has the lowest precision (0.652) due to confusion with adjacent categories, which is expected behavior for ordinal classification problems.

3.1.4 Confusion Matrix Analysis

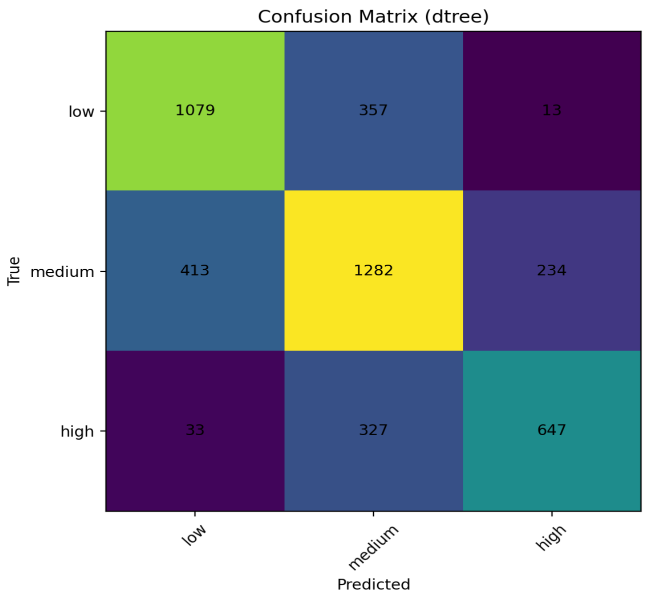

Figure 3.1: Confusion Matrix for Decision Tree Classifier

The confusion matrix reveals that misclassifications predominantly occur between adjacent rating categories. For the low class, 357 samples were misclassified as medium but only 13 as high, demonstrating the ordinal nature of the target. Similarly, high ratings show 327 misclassifications as medium versus only 33 as low. The medium category experiences the most boundary confusion with 413 misclassified as low and 234 as high. This pattern is consistent with ordinal classification challenges.

3.1.5 Feature Importance Analysis

Table 3.3: Top 10 Features by Permutation Importance

#	Feature	Importance	Std Dev
1	YearPublished	0.1451	±0.0034
2	NumWant	0.1250	±0.0071
3	ComWeight	0.0902	±0.0022

4	Cat:War	0.0111	±0.0017
5	NumUserRatings	0.0086	±0.0018

Permutation importance analysis reveals that temporal and community engagement metrics dominate predictive power. YearPublished and NumWant (number of users wanting the game) are the most influential features, together accounting for over 27% of model importance. This suggests that game recency and user demand are strong indicators of rating category. Game complexity (ComWeight) follows as a secondary predictor.

3.2 Regression Task

One regression target was selected: NumOwned (number of users owning the game, log-transformed). Three algorithms were evaluated: Ridge Regression, Random Forest, and Histogram-based Gradient Boosting.

3.2.1 Single Feature Regression

A single linear regression model using NumUserRatings as the sole predictor was first evaluated. This feature showed the highest absolute correlation (0.949) with the log-transformed NumOwned target.

Table 3.4: Single Feature Regression Results

Feature	Correlation	RMSE	MAE	R^2
NumUserRatings	0.949	0.427	0.314	0.903

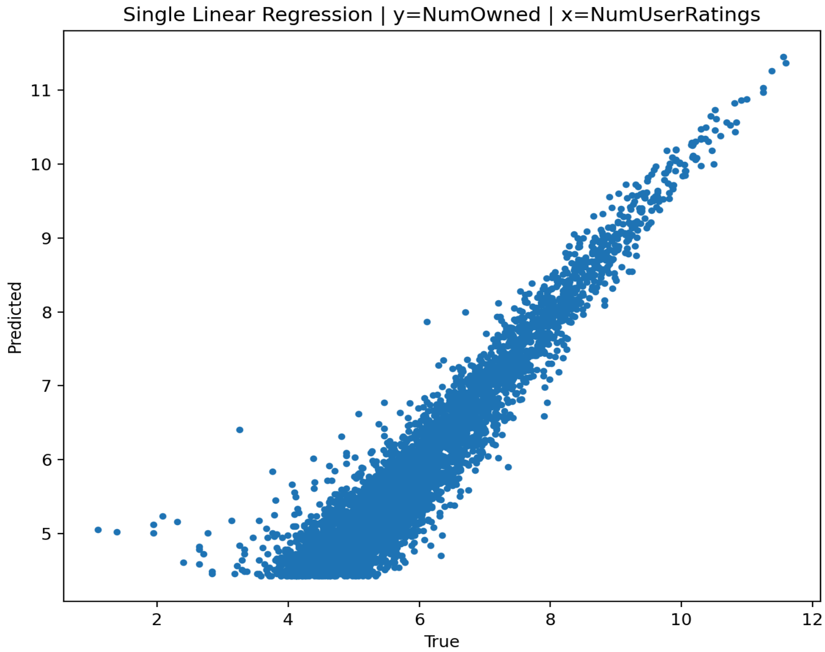

Figure 3.2: Predicted vs True Values for Single Linear Regression

3.2.2 Multiple Regression Model Comparison

Table 3.5: NumOwned Regression Model Comparison

Model	Best Parameters	RMSE	MAE	R^2
HGB	lr=0.1, depth=10	0.350	0.247	0.935
RF	depth=None, leaf=3	0.361	0.254	0.930
Ridge	α=3.0	0.379	0.270	0.923

The NumOwned target was log-transformed to handle its highly skewed distribution. Histogram-based Gradient Boosting achieved the best performance with R^2=0.935, explaining 93.5% of variance in ownership counts. All models perform well (R^2>0.92), but ensemble methods outperform linear Ridge regression.

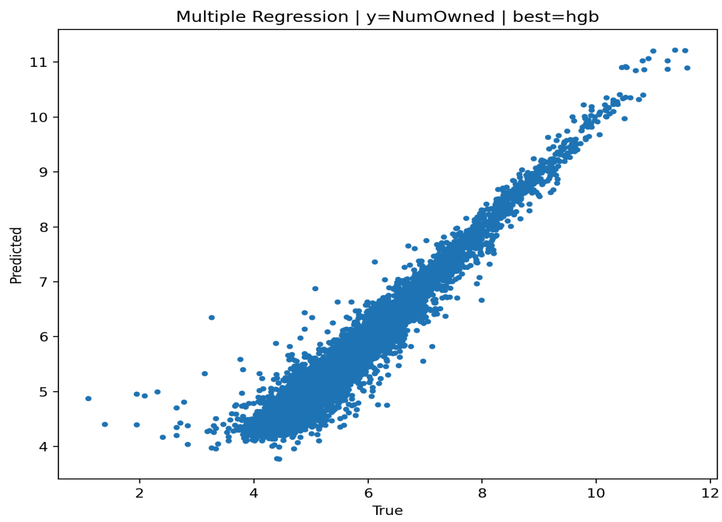

Figure 3.3: Predicted vs True Values for NumOwned (HGB)

3.3 Discussion and Conclusions

3.3.1 Best Models Summary

Table 3.6: Summary of Best Performing Models

Task	Best Model	Primary Metric	Value
Classification (Rating)	Decision Tree	F1-Macro	0.688
Regression (NumOwned)	HGB	R^2	0.935

3.3.2 Key Findings

The Decision Tree classifier outperformed other algorithms for classification, achieving 68.6% accuracy. For classification, this accuracy represents a reasonable result given the inherent difficulty of ordinal three-class classification, where adjacent categories naturally overlap. The confusion patterns confirm the ordinal nature of ratings with minimal extreme misclassifications.

Regression achieved excellent performance with Histogram-based Gradient Boosting (R^2=0.935). The single-feature baseline using NumUserRatings already explained 90.3% of variance, indicating a strong linear relationship between number of ratings and ownership counts.

3.3.3 Limitations and Future Work

The classification task could benefit from ordinal regression approaches that explicitly model the ordered nature of rating categories. Additionally, the strong correlation between NumUserRatings and NumOwned suggests these features may share underlying factors related to game popularity. Future work should explore feature selection techniques to identify a minimal feature set that maintains predictive performance while improving model interpretability.

4. Pattern Mining

This chapter presents the pattern mining analysis performed on the board games dataset containing 21,925 transactions. The objective is to discover frequent itemsets and derive meaningful association rules that reveal hidden relationships among game attributes. The analysis explores how different support and confidence thresholds affect the number of discovered patterns and rules, followed by a detailed examination of the most significant findings.

4.1 Dataset Configuration

The pattern mining was conducted using the following configuration:

- **Total Transactions:** 21,925 board games
- **Binary Columns (11):** NumComments, IsReimplementation, Kickstarted, Cat:Thematic, Cat:Strategy, Cat:War, Cat:Family, Cat:CGS, Cat:Abstract, Cat:Party, Cat:Childrens

- **Quantitative Columns Binned (4 quantiles):** MfgPlaytime, NumOwned, YearPublished, NumUserRatings
- **Target Variable:** Rating (High, Medium, Low)
- **Maximum Itemset Length:** 3
- **Minimum Support Values Tested:** 0.01, 0.02, 0.03
- **Minimum Confidence Values Tested:** 0.30, 0.40, 0.50, 0.60

4.2. Frequent Pattern Extraction

Frequent pattern extraction was performed using the FP-Growth algorithm with varying minimum support thresholds. This analysis reveals how the number of discovered patterns changes with different support levels and identifies the most common attribute combinations in the dataset.

4.2.1 Quantitative Analysis: Patterns vs Minimum Support

Table 1 summarizes the number of frequent itemsets discovered at each minimum support threshold:

Min Support	# Frequent Patterns	% Change
0.01 (1%)	945	—
0.02 (2%)	542	-42.6%
0.03 (3%)	334	-38.4%

Table 4.1: Number of frequent patterns at different support thresholds

The relationship between minimum support and the number of patterns follows an exponential decay pattern. Increasing the support threshold from 1% to 3% reduces the pattern count by approximately 65% (from 945 to 334 patterns). This behavior is expected as higher support thresholds filter out less frequent combinations, retaining only the most prevalent patterns in the dataset.

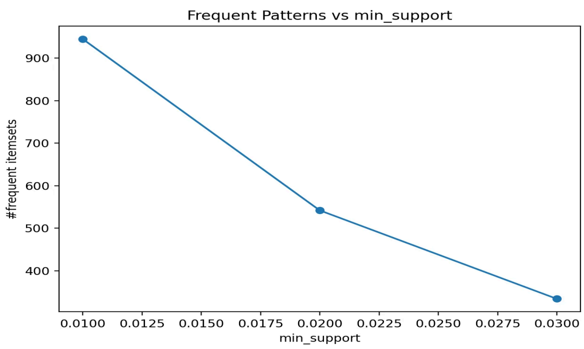

Figure 4.1: Number of frequent patterns as a function of minimum support threshold

4.2.2 Most Frequent Patterns Analysis

At minimum support of 0.02, the top 10 most frequent patterns reveal important characteristics of the board game dataset:

Itemset	Length	Support	Count
Rating=Medium	1	44.0%	9,644
Rating=Low	1	33.0%	7,245
MfgPlaytime=Q2	1	30.9%	6,766
YearPublished=Q3	1	30.2%	6,612
NumOwned=Q4 & NumUserRatings=Q4	2	22.5%	4,940

Table 4.2: Top 5 most frequent patterns at min_support=0.02

Key Observations from Frequent Patterns:

- **Rating Distribution:** Medium-rated games dominate (44%), followed by Low (33%) and High (23%). This indicates most board games receive average ratings.
- **Popularity Correlation:** The pattern {NumOwned=Q4 & NumUserRatings=Q4} with 22.5% support shows a strong correlation between game ownership and rating activity—popular games receive more ratings.

- **Publication Trends:** YearPublished=Q3 (30.2%) suggests more games in the dataset were published in the third quartile of publication years.

4.3. Association Rules Extraction

Association rules were generated from the frequent itemsets at min_support=0.02 using varying confidence thresholds. These rules reveal directional relationships between game attributes.

4.3.1 Quantitative Analysis: Rules vs Minimum Confidence

Table 3 shows how the number of association rules varies with different confidence thresholds:

Min Confidence	# Rules	% Change	Avg Lift
0.30 (30%)	746	—	~1.8
0.40 (40%)	428	-42.6%	~2.2
0.50 (50%)	251	-41.4%	~2.5
0.60 (60%)	162	-35.5%	~2.9

Table 4.3: Number of association rules at different confidence thresholds (min_sup=0.02)

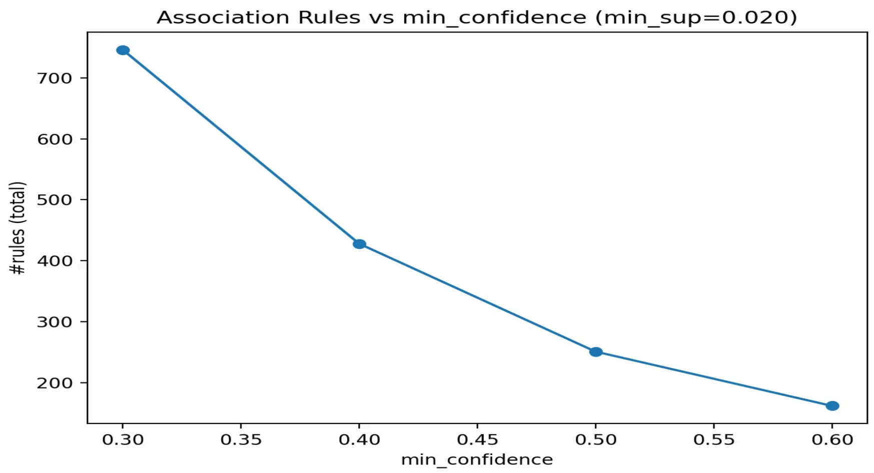

Figure 4.2: Number of association rules as a function of minimum confidence threshold

4.3.2 Rule Quality Distributions

Analyzing the distribution of confidence and lift values provides insights into the quality and significance of the extracted rules. Figures 3 and 4 show histograms for rules extracted at min_support=0.02 and min_confidence=0.40.

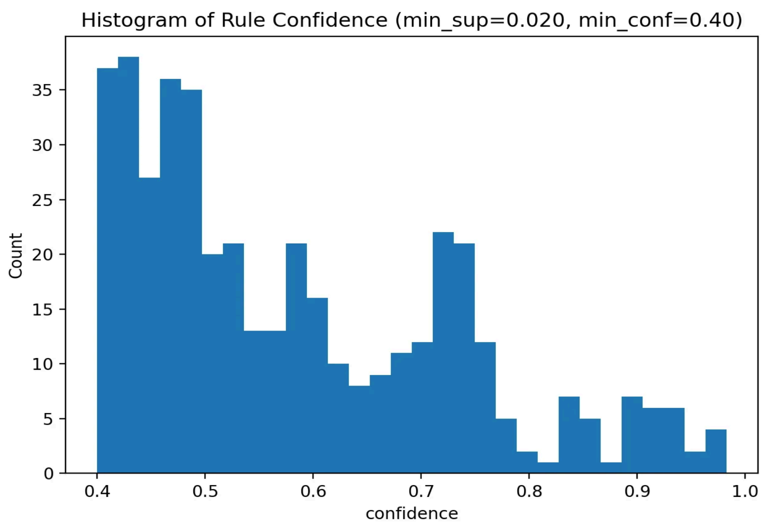

Figure 4.3: Distribution of rule confidence values

The confidence distribution shows a right-skewed pattern with most rules having confidence between 0.40-0.55. A notable secondary peak around 0.70-0.75 indicates a subset of highly reliable rules, while rules with confidence above 0.90 represent the strongest associations in the dataset.

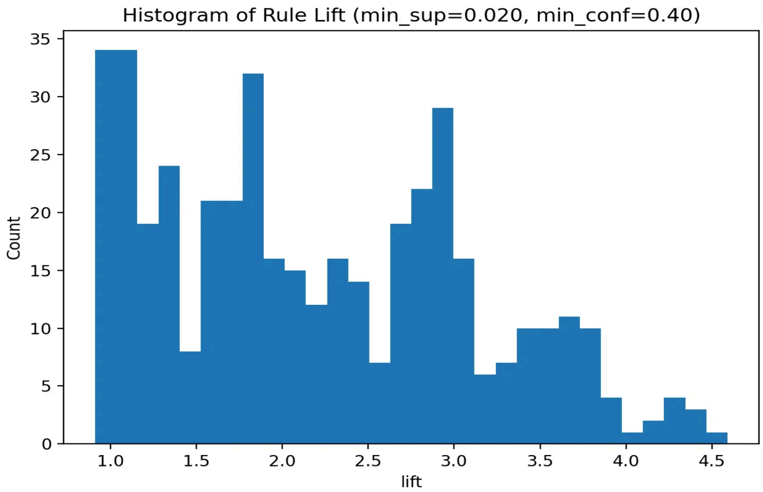

Figure 4.4: Distribution of rule lift values

The lift distribution reveals that most rules have lift values between 1.0-3.5, indicating positive associations (items appear together more often than by chance). Rules with lift > 3.5 represent particularly strong associations, with the highest lift values approaching 4.6.

4.3.3 Top Association Rules by Lift

Table 4 presents the top association rules ranked by lift, which measures how much more likely items appear together compared to random chance:

Rule (LHS → RHS)	Confidence	Support	Lift
MfgPlaytime=Q4 & YearPublished=Q1 → Cat:War=1	73.9%	5.8%	**4.59**
MfgPlaytime=Q2 & NumUserRatings=Q4 → Cat:Family=1	47.2%	3.9%	**4.47**
MfgPlaytime=Q4 & NumUserRatings=Q4 → Cat:Strategy=1	46.7%	2.2%	**4.41**
MfgPlaytime=Q4 & NumOwned=Q2 → Cat:War=1	68.8%	3.7%	**4.27**
Cat:Strategy=1 & NumOwned=Q4 → NumUserRatings=Q4	98.3%	7.3%	**3.93**

Table 4.4: Top 5 association rules by lift at min_support=0.02, min_confidence=0.40

Key Interpretations:

- **War Games & Long Playtime:** Games with long playtime (Q4) published earlier (Q1) are 4.59 times more likely to be war games. This reflects the historical prevalence of complex war simulations.
- **Family Games & Popularity:** Medium playtime games (Q2) with high ratings activity are 4.47 times more likely to be family games, suggesting family games have broader appeal.
- **Strategy Games Engagement:** Popular strategy games (highest ownership quartile) have 98.3% probability of high rating activity, showing engaged player communities.

4.4. Exploiting Association Rules for Rating Prediction

To demonstrate the practical utility of the extracted association rules, we implemented a rule-based classifier for predicting game ratings. Rules with Rating in the consequent were used to predict High, Medium, or Low ratings for the test set.

4.4.1 Methodology

1. Extract all rules where the consequent is Rating=High, Rating=Medium, or Rating=Low
2. For each test instance, find all matching rules (where antecedent conditions are satisfied)
3. Select the rule with highest confidence to make the prediction
4. If no rules match, predict the majority class (Rating=Medium)

4.4.2 Prediction Results

The rule-based classifier was evaluated on a 20% test split with the following results:

Metric	Value
Accuracy	**51.38%**
Macro F1-Score	**50.01%**

Table 4.5: Rule-based classifier performance metrics

Confusion Matrix:

True \ Pred	High	Low	Medium
High	715	158	100
Low	160	1123	204
Medium	773	737	415

Table 4.6: Confusion matrix for rule-based rating prediction

Analysis:

The rule-based classifier achieves 51.38% accuracy, which is above the baseline of 44% (majority class prediction). The confusion matrix reveals:

- **Low Rating Prediction:** Best performance with 1,123 correct predictions out of 1,487 true Low ratings (75.5% recall)
- **High Rating Prediction:** Good performance with 715 correct predictions (73.5% recall)
- **Medium Rating Prediction:** Challenging with only 415 correct predictions (21.5% recall), often misclassified as High or Low

4.5. Conclusion

The pattern mining analysis of the board games dataset revealed several meaningful insights:

- **Pattern Discovery:** 542 frequent patterns were discovered at min_support=0.02, with the number decreasing exponentially as support threshold increases.
- **Strong Associations:** 428 association rules were extracted at min_confidence=0.40, revealing relationships between playtime, publication year, game categories, and ratings.
- **Domain Insights:** War games are strongly associated with long playtimes and older publication dates. Strategy and Family games show high engagement patterns.
- **Practical Application:** The rule-based classifier achieved 51.38% accuracy for rating prediction, demonstrating the utility of association rules beyond descriptive analysis.

The chosen configuration of min_support=0.02 and min_confidence=0.40 provides a good balance between discovering meaningful patterns and filtering noise, producing interpretable rules with lift values indicating genuine associations rather than random co-occurrences.